

Azmi-Salah Bousedra

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EDUCATION

Concordia University

Bachelor of Software Engineering (BEng)

Montreal, QC

Sept. 2024 – May 2028

Champlain College

Health Science Collegial Diploma (DEC)

Saint-Lambert, QC

Aug. 2021 – Apr. 2023

EXPERIENCE

Firmware R&D Intern

Sep. 2025 – Dec. 2025

PixMob

Montreal, QC

- Designed and deployed a complete **CI/CD pipeline** that compiles, documents, validates, and auto-flashes firmware to the correct target device using **GitLab CI**, reducing manual effort by **70 percent** while ensuring production-grade traceability.
- Built a **unified test automation framework** with custom **firmware** for the devices, a **UART** trace parser, and an **XML** report generator, streaming real-time hardware results into **GitLab CI** and reducing testing time by **30 percent**.
- Architected a parallel **firmware flashing system** leveraging **embedded firmware** and relay-controlled I/O to orchestrate **multi-device** programming, enabling scalable production workflows and reliable device provisioning.

PROJECTS

Ironify (ConUHacks) | Python, FastAPI, TensorFlow, Gemini API, MongoDB

Jan. 2026

- Engineered a high-performance asynchronous **FastAPI** backend served by **Uvicorn**, utilizing **Motor** for non-blocking **MongoDB** transactions and **python-multipart** for efficient file ingestion.
- Integrated **Google Gemini** to build an autonomous threat analyst that parses raw logs into structured JSON, calculating risk scores (0-100) and generating code-level remediation patches which can be executed.
- Developed an unsupervised **Deep Neural Network (DNN)** using **TensorFlow** Autoencoders to conduct forensic audits, identifying anomalies in historical datasets based on reconstruction loss (MSE).

Comments4Me | JavaScript, Node.js, Express, Prism.js, Gemini API

Feb. 2025 – May 2025

- Built an **AI-powered** web application using the **Gemini API** to analyze source code and generate automated documentation comments, improving developer workflow and code maintainability.
- Implemented **Prism.js** for real-time **syntax highlighting** and code block rendering, delivering a polished, developer-centric interface.
- Integrated **front-end** and **back-end** pipelines to support **real-time** code parsing, interactive user input, and dynamic comment visualization.

TuneWear | C++, ESP32, LasFM API

Jul. 2025 – Sep. 2025

- Designed and built an **ESP32-powered** wearable clip with **real-time** track display sourced from streaming services, enabling live play-along sessions on instruments such as guitars and pianos.
- Developed a customizable **Blynk** mobile dashboard using **WiFi-based APIs** to control display modes, configure visual styles, and monitor **real-time** device status.
- Engineered the full hardware assembly by selecting an **ESP32-T5** board, integrating a **LiPo battery**, and adapting a **3D-printed** enclosure to deliver a compact wearable without requiring a custom **PCB**.

AfterLab | ExtendScript, JavaScript, XML

Dec. 2025 – Present

- Building an automation tool for **Adobe After Effects** to reduce manual editing time by scripting common motion design workflows.
- Implementing **ExtendScript** to enable **CRUD operations** on compositions and layers (create, read, update, delete), delivering reliable project-level automation.
- Building a responsive product landing page using **React** and **GSAP**, implementing animated UI flows to support AfterLab commercialization and user acquisition.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, HTML/CSS, Bash

Frameworks, APIs & Libraries: FastAPI, Node.js, Express, React, TensorFlow, Gemini API, Prism.js, GSAP

DevOps & Tooling: GitLab CI, Git, Docker, Linux, Vultr, Visual Studio Code, JetBrains IDEs

Databases: MongoDB